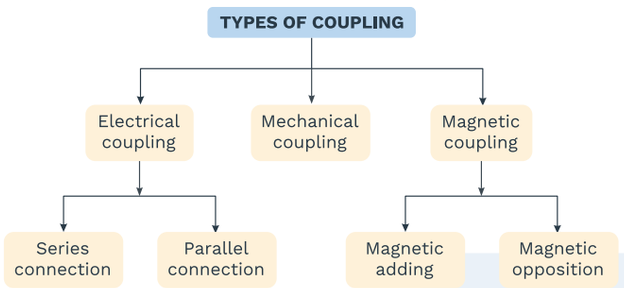
9 Magnetically Coupled Circuit

INTRODUCTION

Coupling

When the two systems or devices are connected together and they work as a single system, then the systems are said to be as coupled.



Electrical coupling:

When an electrical quantity such as voltage or current couples the two systems together, then it is said to be as electrical coupling systems.

Example: All the household or the domestic devices are connected in parallel, so they are termed as electrically coupled.

Mechanical coupling:

When the two systems are connected together in such a manner that they move together like a motor and a fan in windmill, motor and blades of ceiling fans are termed as mechanically coupled.

Magnetic coupling:

When the two parts of a system have magnetic field in common, or we can say there is a magnetic field that passes through both the parts of a system, then the parts are termed as magnetically coupled.

For example: Transformers, dc machines, induction machines.

SERIES MAGNETIC CIRCUIT

The system in which a common flux exists throughout the circuit is termed as series magnetic circuit.

Definitions

Coupling:
Coupling is defined as a desirable or undesirable transfer of energy from one medium to another medium.

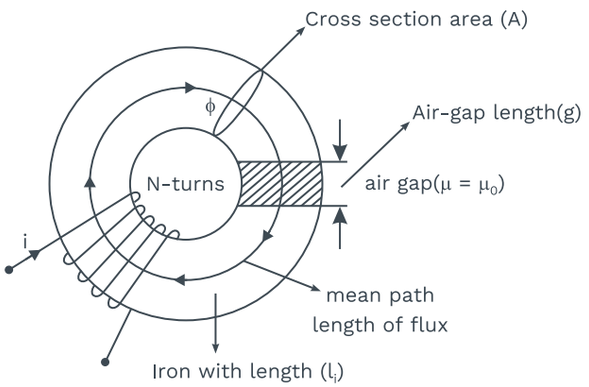


Fig. 9.1 Series Magnetic Circuit

From the above figure, common flux (ϕ) exists in both iron and air.

Permeability of iron, $S_{iron} = \frac{l_i}{\mu_0 \mu_r A}$

Permeability of air-gap, $S_{gap} = \frac{g}{\mu_0 A}$

Flux is the same i.e., ϕ = same

$S = S_{iron} + S_{gap}$

PARALLEL MAGNETIC CIRCUIT

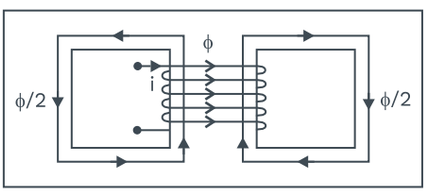


Fig. 9.2 Parallel Magnetic Circuit

- In the above circuit, the flux (ϕ) due to the coil gets divided into two parts, and those two parts are said to be connected parallelly in a magnetic circuit.

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

"Whenever there is a rate of change of flux with respect to time through the coil, an emf is induced."

Definitions

Inductance:
The property of the electric conductor by which the change in current produces an electromotive force (emf) is known as inductance.

Rack Your Brain

Does inductance depends on the frequency and current?

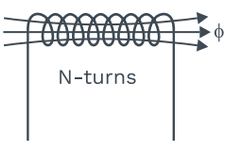


Fig. 9.3 Flux Through the Coil Having N Number of Turns

- Magnetic flux through each turn = ϕ

  Total flux linking the coil, flux linkage, $\Psi = N\phi$
- From Faraday's law, the emf induced = $\frac{d\Psi}{dt}$

$$e = \frac{d}{dt}(N\phi)$$

$$e = N\frac{d\phi}{dt}$$

LENZ'S LAW

- Induced emf opposes the cause of induction.
- The current flow due to the induced emf will oppose the change of flux and will try to maintain flux in the circuit i.e. ϕ = constant.

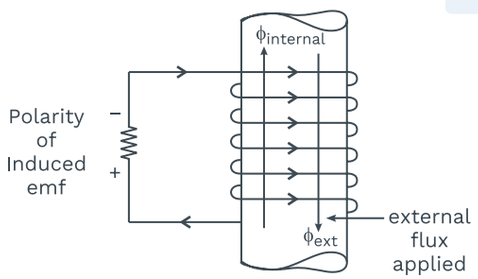


Fig. 9.4 Illustration of the Polarity of an Induced emf

Steps to calculate the polarity of an induced emf:

- Consider a ϕ-coil opposite to the external emf.
- Based on ϕ-coil, determine the current direction in a coil.
- Connect a resistor across a coil and determine the voltage polarity across a resistor, and the same polarity exists for induced emf.

SELF INDUCTANCE

If current 'i' flows in the coil,

Magnetic field, $H \propto i$

Flux density, $B = \mu H \propto i$

Flux, $\phi = BA \propto i$

Flux linkage, $\Psi = N\phi \propto i$

$\Psi \propto i$

$\Psi = Li$

Where L is the proportionality constant or self-inductance (Henry, H)

Definitions

Self-inductance:
The property of current-carrying coil by virtue of which it opposes the change of current in the circuit by inducing a current in itself is termed as self-inductance.

MUTUAL INDUCTANCE

Flux produced by a coil induces an electromotive force (emf) in other coils.

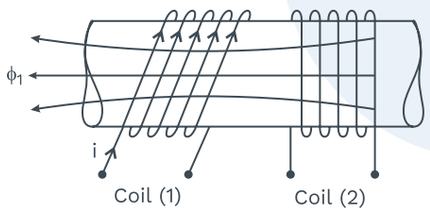


Fig. 9.5 Induced emf in Coil 2) Due to Coil 1)

Definitions

Mutual inductance:
When a current passes through a coil (primary) changes, the magnetic flux of the neighbouring coil (secondary) also changes. Thus, an emf is produced in the neighbouring coil. This phenomenon is termed as mutual inductance.

- The flux of coil 1) links with the other coil 2), as both the coils are on the same core, so emf of coil 2) depends on the current of coil 1).
  If emf of coil 1) is $F_1 = N_1 i_1$

  So, flux $\phi = \frac{F_1}{s} = \frac{N_1 i_1}{s}$

  Flux linkage in coil 2) due to coil 1) is:

  $$\Psi_{12} = N_2 \phi_1 = \frac{N_1 N_2 i_1}{s}$$

  $$\frac{\Psi_{12}}{i_1} = \text{Mutual inductance}$$

  $$M_{12} = \frac{\Psi_{12}}{i_1} = \frac{N_1 N_2}{s}$$

  Similarly, if we consider current is coil 2) and flux in coil 1) is:

Rack Your Brain

Does the mutual inductance depends on the current?

$$\Psi_{21} = \frac{N_1 N_2 i_2}{s}$$

$$M_{21} = \frac{\Psi_{21}}{i_2}$$

$$M_{21} = \frac{N_1 N_2}{s}$$

or

$$\delta = \frac{s}{\mu A}$$

$$M = \frac{\mu A N_1 N_2}{\ell}$$

The total energy stored in a circuit is:

$$E_{total} = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 \pm M i_1 i_2$$

(+) → Additive coupling: Mutual adding flux
(–) → Subtractive coupling: Mutual opposite flux

Rack Your Brain

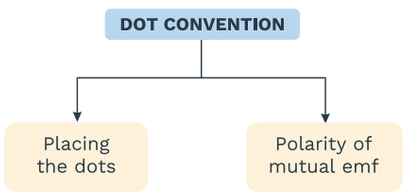
Why does the mutual inductance happen?

DOT CONVENTION

- Dots are placed at different terminals of the coil in order to determine the polarity of an induced mutual emf.
- For resistance, inductance and capacitance, we determine the polarity of voltage by following passive element sign convention, but to determine the polarity of mutual emf we have to follow the dot convention.

Grey Matter Alert!

If the current entering through the dotted terminal of a coil, then the reference polarity of mutual voltage at the dotted terminal of the second coil is positive.



Placing the dots:
Steps for placing the dots are:

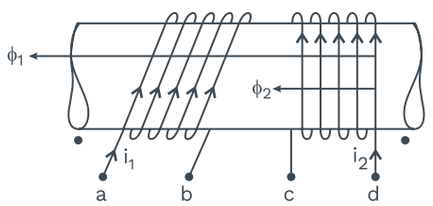


Fig. 9.6 Illustrating the Placement of Dots in the Circuit

Step I: Assume a current in one of the coils and place a dot at the terminal where the current enters the coil.

Step II: Determine the magnetic field direction based on the current direction assumed.

Step III: Assume the additive of the second coil in same direction as the first coil.

Step IV: Obtain the direction of the current in the second coil depending on the magnetic field of the second coil.

Step V: Place the dot on that terminal of the second coil from where the current enters.

For example:

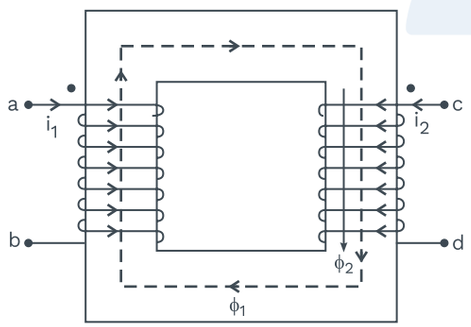


Fig. 9.7 a) By Additive Coupling

Note:

While placing the dot, it doesn't matter if the coils are connected to each other, or if there is a source connected to one of the coils on there is a lead connected to one of the coils.

Grey Matter Alert!

If the current is leaving at the dotted terminal of a coil, the reference polarity of the mutual voltage at the dotted terminal of the second coil is negative.

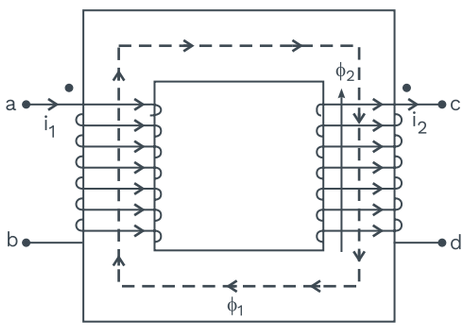


Fig. 9.7 b): By Subtractive Coupling

- In figure 9.7 b), we assume the flux ϕ_2 is opposite to the flux ϕ_1.
- We place a dot on that terminal of the second coil where the current i_2 leaves the coil 2.

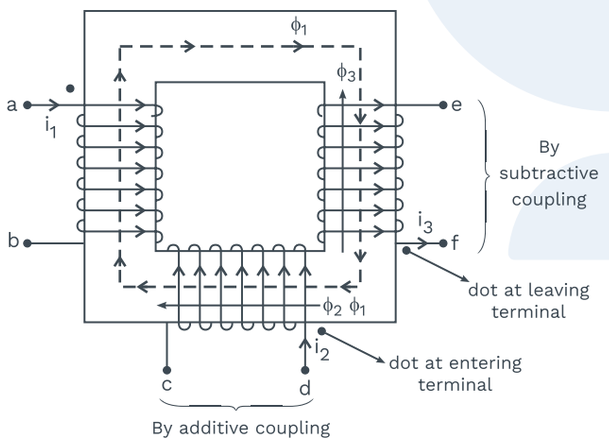


Fig. 9.8 Dot Convention in Both the Coils

POLARITY OF MUTUAL EMF

- The practical use of dot convention is to determine the polarity of mutually induced emf.

  Rule 1: If both the currents in both the coils either enters or leaves the dot, then it is additive polarity. In such a case, the polarity of mutual emf is as per passive element.

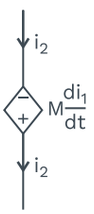
  Thus, the current enters the positive terminal of mutual emf.

Grey Matter Alert!

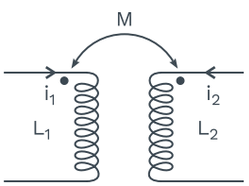
It is important to know that mutual coupling exists only when the inductors are placed close together and input excitation is time-varying.



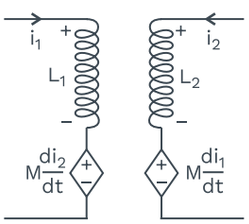
Rule 2: If the current enters the dot in one coil and leaves the dot in another coil then it is termed as subtractive polarity, then we follow the sign the convention of the active element. Thus, the current leaves the positive terminal.



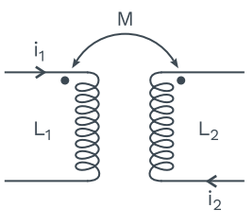
Case I:



Here, the current enters the dot on both sides. Therefore, it is an additive coupling.



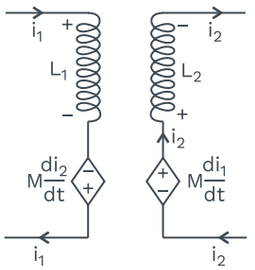
Case II:



Rack Your Brain

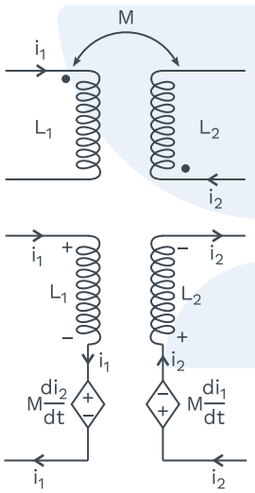
Can mutual inductance be negative?

In this case, the current enters the dot on i_1 and leaves the dot on i_2. Therefore, it is a subtractive coupling.



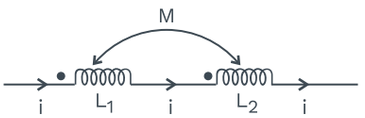
Case III:

Here, both the currents enter the dot, so it is an additive coupling.



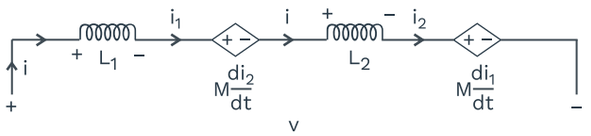
SERIES CONNECTION

Case I: Additive coupling:



Current enters both the dots

$\therefore \quad i = i_1 = i_2$



By applying KVL-

$$v = L_1\frac{di_1}{dt} + M\frac{di_2}{dt} + L_2\frac{di_2}{dt} + M\frac{di_1}{dt}$$

$\therefore \quad i = i_1 = i_2$

$$v = L_1\frac{di}{dt} + M\frac{di}{dt} + L_2\frac{di}{dt} + M\frac{di}{dt}$$

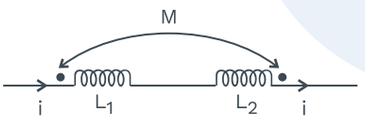
$$v = \frac{di}{dt}[L_1 + M + L_2 + M]$$

$$v = \frac{di}{dt}[L_1 + L_2 + 2M]$$

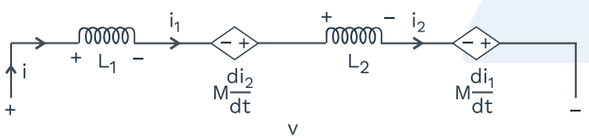
$$v = \frac{di}{dt}\cdot L_{eq}$$

where $L_{eq} = L_1 + L_2 + 2M$

Case II: Subtractive coupling:



Current enters in one dot and leaves the other.



Applying KVL in the network:

$$v = L_1\frac{di_1}{dt} - M\frac{di_2}{dt} + L_2\frac{di_2}{dt} - M\frac{di}{dt}$$

$\therefore \quad i = i_1 = i_2$

$$v = L_1\frac{di}{dt} - M\frac{di}{dt} + L_2\frac{di}{dt} - M\frac{di}{dt}$$

$$v = \frac{di}{dt}[L_1 - M + L_2 - M]$$

$$v = \frac{di}{dt}[L_1 + L_2 - 2M]$$

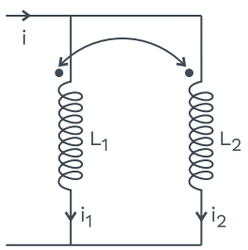
$$\frac{v}{\frac{di}{dt}} = L_{eq} = L_1 + L_2 - 2M$$

Previous Years' Question

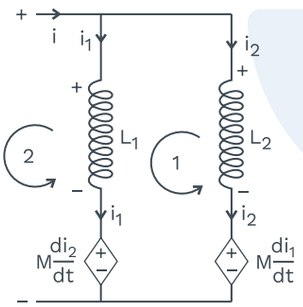
Two identical coils of negligible resistance when connected in series across a 200 V, 50 Hz. The source draws a current of 10 A. When the terminals of one of the coils are reversed, then current drawn is 8 A. The coefficient of coupling between the two coils is _____.

PARALLEL CONNECTION

Case I: Additive polarity:



Since both the currents are entering the dot, so it is an additive coupling.



On applying KVL in the loop 1), we get:

$$-L_1\frac{di_1}{dt}-M\frac{di_2}{dt}+M\frac{di_1}{dt}+L_2\frac{di_2}{dt}=0$$

$$(L_1-M)\frac{di_1}{dt}=(L_2-M)\frac{di_2}{dt}$$

$$\frac{di_2}{dt}=\frac{(L_1-M)}{(L_2-M)}\frac{di_1}{dt}$$

$$i = i_1 + i_2$$

$$\frac{di}{dt}=\frac{di_1}{dt}+\frac{di_2}{dt}$$

$$\frac{di}{dt}=\frac{di_1}{dt}+\frac{di_1}{dt}\left[\frac{(L_1-M)}{(L_2-M)}\right]$$

$$\frac{di}{dt}=\frac{di_1}{dt}\left[1+\frac{L_1-M}{L_2-M}\right]$$

$$\frac{di}{dt}=\frac{di_1}{dt}\left[\frac{L_2-M+L_1-M}{L_2-M}\right]$$

$$\frac{di}{dt} = \frac{di_1}{dt}\left[\frac{L_1 + L_2 - 2M}{L_2 - M}\right] \quad \text{...i)}$$

Now, applying KVL in loop 2):

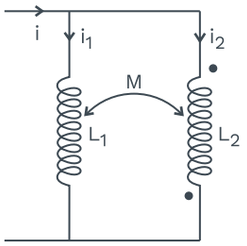
$$v = L_1\frac{di_1}{dt} + M\frac{di_2}{dt}$$

$$v = \left[L_1 + M\frac{(L_1 - M)}{(L_2 - M)}\right]\frac{di_1}{dt}$$

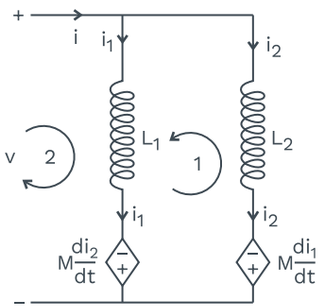
$$L_{eq} = \frac{v}{\frac{di}{dt}} = \frac{L_1L_2 - ML_1 + ML_1 - M^2}{L_1 + L_2 - 2M}$$

$$L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$$

Case II: Subtractive polarity:



One current enters the dot while the other current leaves the dot. Thus, it is a subtractive coupling.

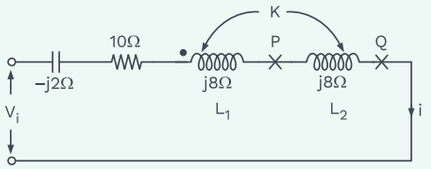


Applying KVL in loop 1):

$$-L_1\frac{di_1}{dt} + M\frac{di_2}{dt} - M\frac{di_1}{dt} + L_2\frac{di_2}{dt} = 0$$

Previous Years' Question

In the circuit shown in figure, it is found that the input AC voltage (V_i) and current 'i' are in phase. The coupling coefficient is $K = \frac{M}{\sqrt{L_1 L_2}}$, where M is the mutual inductance between the two coils. The value of K and the dot polarity of the coil P-Q are.



a) K = 0.25 and dot at P
b) K = 0.5 and dot at P
c) K = 0.25 and dot at Q
d) K = 0.5 and dot at Q

$$\frac{di_2}{dt} + [M + L_2] = [M + L_1]\frac{di_1}{dt}$$

$$\frac{di_2}{dt} + \left[\frac{L_1 + M}{L_2 + M}\right]\frac{di_1}{dt}$$

$$i = i_1 + i_2$$

$$\frac{di}{dt} = \frac{di_1}{dt} + \left[\frac{L_1 + M}{L_2 + M}\right]\frac{di_1}{dt}$$

$$\frac{di}{dt} = \frac{di_1}{dt}\left[1 + \frac{(L_1 + M)}{(L_2 + M)}\right]$$

$$\frac{di}{dt} = \frac{di_1}{dt}\left[\frac{L_2 + M + L_1 + M}{L_2 + M}\right]$$

$$\frac{di}{dt} = \frac{di_1}{dt}\left[\frac{L_1 + L_2 + 2M}{L_2 + M}\right]$$

By applying KVL in loop-2,

$$v = L_1\frac{di_1}{dt} - M\frac{di_2}{dt}$$

$$v = \left[L_1 - M\frac{(L_1 + M)}{(L_2 + M)}\right]\frac{di_1}{dt}$$

$$L_{eq} = \frac{v}{\frac{di}{dt}}$$

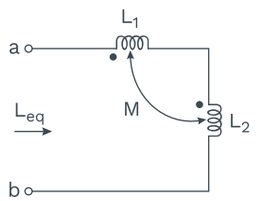
$$L_{eq} = \frac{L_1 + L_2 + L_1M - L_1M - M^2}{L_1 + L_2 + 2M}$$

$$L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$$

- For additive polarity, the inductance must be higher, and the denominator must be smaller which gives $(L_1 + L_2 - 2M)$.
- For subtractive polarity, the inductance must be smaller, and the denominator must be higher which gives $(L_1 + L_2 + 2M)$.

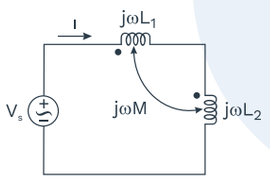
Example 1:

For the coupled coils shown in the figure, the equivalent inductance L_{eq} at terminals a – b is:



Sol:

Consider the circuit in phasor domain



$$V_s = (j\omega L_1)I + (j\omega M)I + (j\omega L_2)I + (j\omega M)I$$

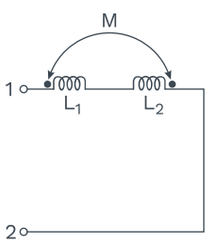
$$V_s = (L_1 + L_2 + 2M)\, j\omega I$$

$$\frac{V_s}{I} = j\omega L_{eq} = j\omega(L_1 + L_2 + 2M)$$

$$L_{eq} = L_1 + L_2 + 2M$$

Example 2:

Determine the equivalent inductance (L_{eq}) measured between the terminals 1 and 2 for the circuit shown in the figure:



Sol:

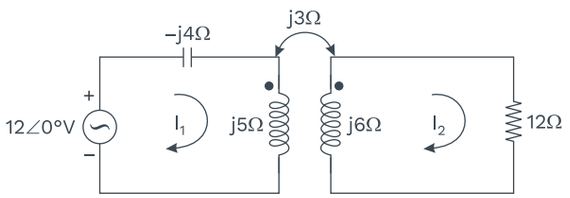
In the given figure, the current is entering the dotted terminal of the first coil (L_1), and it exits from the dotted terminal of the second coil (L_2).

The above case is a series of opposing connection and the mutual inductance is:

$L_{eq} = L_1 + L_2 - 2M$

Example 3:

Obtain the phasor current I_1 and I_2 for the circuit shown in the figure below.



Sol:

Applying KVL in coil-1, we get:

$12 = (-j4 + j5)\, I_1 - j3I_2$

or $jI_1 - j3I_2 = 12$

$jI_1 - j3I_2 = 12$ ...(i)

Now applying KVL in coil-2, we get:

$-j3I_1 + (12 + j6)\, I_2 = 0$

$$I_1 = \frac{(12 + j6)I_2}{j3} = (2 - j4)I_2$$ ...(ii)

Substituting this in equation i), we get:

$(j2 + 4 - j3)\, I_2 = (4 - j)\, I_2 = 12$

Or $I_2 = 2.91 \angle 14.04°$ Amp ...(iii)

From equation ii) and iii),

$I_1 = (2 - j4)\, I_2$

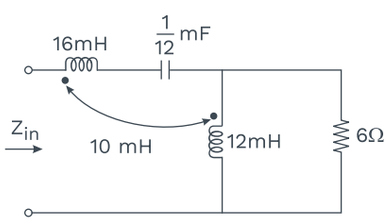
$I_1 = (2 - j4)\, I_2$

$I_1 = (4.472 \angle -63.43°)\, (2.91 \angle 14.04°)$

$I_1 = 13.01 \angle -49.39°$ A ...(iv)

Example 4:

The input impedance Z_{in} of the network shown in figure, at a frequency $\omega = 1$ krad/sec is:



Sol:

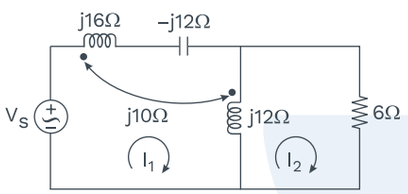
Consider the circuit in phasor domain

$16mH \Rightarrow j\omega L_1 = j(1000)(16 \times 10^{-3}) = j16\Omega$

$12mH \Rightarrow j\omega L_2 = j(1000)(12 \times 10^{-3}) = j12\Omega$

$10mH \Rightarrow j\omega M = j(1000)(10 \times 10^{-3}) = j10\Omega$

$$\frac{1}{12}mF \Rightarrow \frac{1}{j\omega C} = \frac{1}{j(1000)\left(\frac{1}{12} \times 10^{-3}\right)} = -j12\Omega$$



Writing mesh equation

Mesh 1:

$$V_s - j16I_1 - (-j12)I_1 - j12(I_1 - I_2) - \underbrace{j10(I_1 - I_2) - j10I_1}_{\text{Due to mutual inductance}} = 0$$

$V_s = j16I_1 - j12I_1 + j12(I_1 - I_2) + j10(I_1 - I_2) + j10I_1$

$V_s = I_1(j16 - j12 + j12 + j10 + j10) - I_2(j12 + j10)$

$V_s = j36I_1 - j22I_2$ .....i)

Mesh 2:

$-j12(I_2 - I_1) - 6I_2 - (-j10I_1) = 0$

$j12(I_2 - I_1) + 6I_2 - j10I_1 = 0$

$-I_1(j12 + j10) + I_2(j12 + 6) = 0$

$j22I_1 = I_2(6 + j12)$

$$I_2 = \frac{j22}{6 + j12}I_1 = 1.64\angle 26.5°\, I_1$$

Substituting I_2 into equation i)

$V_s = j36I_1 - j22(1.64\angle 26.5°)I_1$

$$Z_{in} = \frac{V_s}{I_1} = j36 - (-16 + j32.2)$$

$Z_{in} = 16 + j3.8\Omega$

COUPLING COEFFICIENT

Fraction of total flux produced by a coil that acts as a mutual flux.

$$K = \frac{\phi_m}{\phi_{total}}$$

$$\text{Coefficient of coupling} = \frac{M}{\sqrt{L_1 L_2}}$$

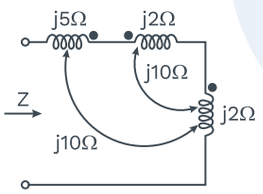
If K < 1, under coupling (most practical)
If K = 1, ideal coupling
If K > 1, order coupling

Definitions

Coupling coefficient:
Fraction of magnetic flux produced by the current in one coil that links with a nearby or neighbouring coil. It is denoted by K.

Example 5:

Determine the impedance in the given figure:



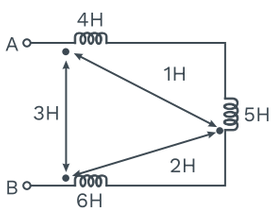
Sol:

In the above figure, j5Ω and j2Ω are in series opposing connection, as the current is entering from the dotted terminal of the j2Ω coil and leaving from the dotted terminal of the j5Ω coil. On the other hand, j2Ω and j2Ω are in series adding a connection, since the current is entering the dotted terminal of both the coil.

Thus, Z = j5 + j2 + j2 – 2(j10) + 2(j10) = j9Ω

Example 6:

The effective inductance of the circuit across the terminals A and B for the figure shown below.



Rack Your Brain

Why the coefficient of coupling in a magnetic circuit is not more than one?

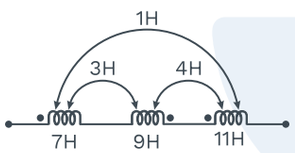
Sol:

According to the positions of the dots (direction of currents), the equivalent inductance of the circuit is obtained as:

$$L_{AB} = L_1 + L_2 + L_3 - 2M_1 + 2M_2 - 2M_3$$
$$= 4 + 5 + 6 - 2(1) + 2(2) - 2(3)$$
$$= 15 - 2 + 4 - 6$$
$$L_{AB} = 11H$$

Example 7:

Determine the equivalent inductance of the circuit shown in the figure?



Sol:

For coil 1,

$$L_{1(eq)} = L_1 - M_{12} + M_{13}$$
$$= 7 - 3 + 1$$
$$L_{1(eq)} = 5H \ (M_{12} = 3H, M_{13} = 1H)$$

For coil 2,

$$L_{2(eq)} = L_2 - M_{21} - M_{23}$$
$$= 9 - 3 - 4$$
$$L_{2(eq)} = 2H \ (M_{23} = 4H)$$

For coil 3,

$$L_{3(eq)} = L_3 + M_{31} - M_{32}$$
$$= 11 + 1 - 4H$$
$$L_{3(eq)} = 8H$$
$$L_{eq} = L_{1(eq)} + L_{2(eq)} + L_{3(eq)}$$
$$= 5 + 2 + 8$$
$$L_{eq} = 15 \ H$$

Alternative method:

$$L_{eq} = L_1 + L_2 + L_3 - 2M_{12} - 2M_{23} + 2M_{13}$$
$$= 7 + 9 + 11 - 2(3) - 2(4) + 2(1)$$
$$= 27 - 6 - 8 + 2$$
$$L_{eq} = 15H$$

TRANSFORMER

- A transformer is a set of two coils that are magnetically coupled as they are wound on the same core but have a different number of turns.

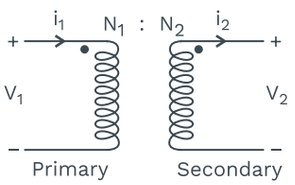


Fig. 9.9 Transformer

Definitions

Transformer:
An electrical device which is designed on the basis of the concept of magnetic coupling is known as the transformer. It uses magnetically coupled coil to transfer energy from one circuit to another.

Assumptions of an ideal transformer:

- Perfect coupling (K = 1).
- No resistance of the coil.
- Relative permeability of the core, $\mu_r \rightarrow \infty$.
- No iron losses.

Rule for an ideal transformer:

Rule-1: Polarity of voltage at both the dotted terminals are identical.

Rule-2: Current enters the dot in one winding and leaves the dot in another winding (subtractive polarity).

For an ideal transformer:

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

From the above equation,

$$V_2 I_2 = V_1 I_1$$

So, apparent power is the same on both sides.

THE IDEAL TRANSFORMER

The transformer with perfect coupling (K = 1) is termed an ideal transformer. The coil on the input side is referred to as the primary coil. The coil on the output side of a transformer is referred to as a secondary coil. The primary coil has N_1 turns, and the secondary has N_2 turns. So, the turns ratio or the transformation ratio is:

$$n = \frac{N_2}{N_1} \qquad ...(1)$$

Definitions

Ideal transformer:
A transformer is considered as an ideal transformer if it is free from all the losses. The primary and the secondary coils of a transformer must have infinite self-inductance.

VOLTAGE RATIO

When a sinusoidal voltage is applied to the primary winding, the same magnetic flux ϕ goes through both windings.

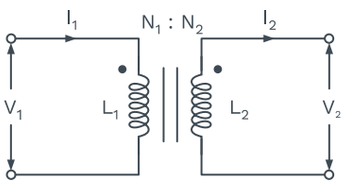


Fig. 9.10 An Ideal Transformer

According to Faraday's law, the voltage across the primary winding is,

$$V_1(t) = N_1 \frac{d\phi}{dt} \quad ...(2)$$

While that across the secondary winding is,

$$V_2(t) = N_2 \frac{d\phi}{dt} \quad ...(3)$$

Dividing equation 3) by equation 2), we get:

$$\frac{V_2(t)}{V_1(t)} = \frac{N_2}{N_1} = n$$

Where 'n' is the transformation ratio. In terms of phasor voltage V_1 and V_2 equations may be written as.

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = n \quad ...(4)$$

"The ratio of secondary voltage to the primary voltage is equal to the ratio of the number of turns in secondary to the number of turns in the primary."

Note:

- If the turn ratio is greater than 1 i.e., n > 1, the output voltage is greater than the input voltage ($V_2 > V_1$), and the transformer is called a step-up transformer.
- If the turn ratio is less than 1 i.e., n < 1, then the output voltage is less than the input voltage ($V_2 < V_1$), and the transformer is called a step-down transformer.

Rack Your Brain

Why source current is zero in ideal transformer under no load condition?

Grey Matter Alert!

The transformer is said to be step-up transformer if secondary voltage is more than its primary voltage.

CURRENT RATIO

For an ideal transformer, the supplied energy to the primary winding should be equal to the energy absorbed by the secondary winding, which implies that:

$$V_1(t)\, I_1(t) = V_2(t)\, I_2(t) \qquad ...(5)$$

In phasor form equation 5) becomes:

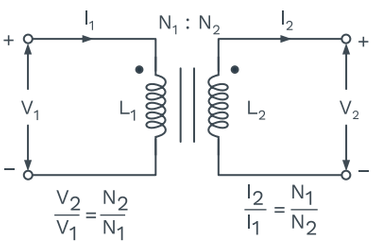
$$\frac{I_1}{I_2} = \frac{V_2}{V_1} = n$$

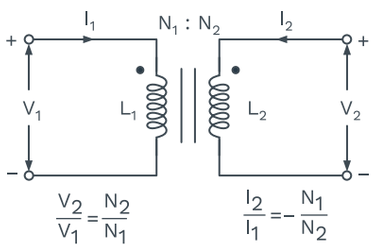
or $$\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n} \qquad ...(6)$$

Note:

- If V_1 and V_2 are both positive or both negative at the dotted terminals, use +n in equation 4) otherwise, use –n.
- If I_1 and I_2 both enter into or both leave the dotted terminal use –n in equation 6) otherwise, use +n.

The above points are explained for the circuit shown below.





Rack Your Brain

What will happen if two transformers are connected in parallel with the wrong polarity?

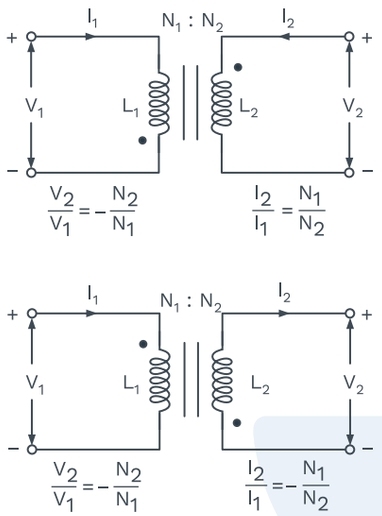
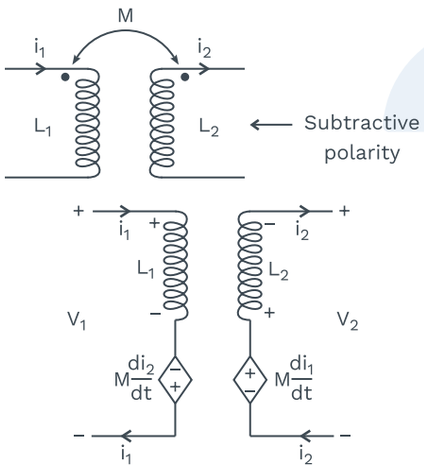


Fig. 9.11: Circuits Illustrating the Voltage Polarities and the Current Directions in an Ideal Transformer

T-MODEL OF TRANSFORMERS



Applying KVL in the circuit:

$$V_1 = L_1 \frac{di_1}{dt} - M\frac{di_2}{dt}$$

$$V_2 = M\frac{di_1}{dt} - L_2\frac{di_2}{dt}$$

$$V_1 = (L_1 - M)\frac{di_1}{dt} + M\frac{di_1}{dt} - M\frac{di_2}{dt}$$

Grey Matter Alert!

The transformer is said to be step-down transformer if primary voltage is more than its secondary voltage.

Previous Years' Question

Given two coupled inductors L_1 and L_2 their mutual inductance M satisfies

a) $M = \sqrt{L_1^2 + L_2^2}$

b) $M > \frac{(L_1 + L_2)}{2}$

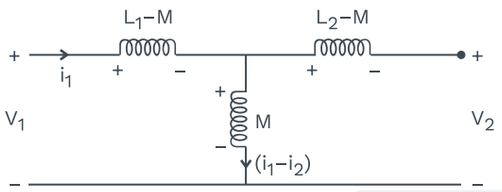
c) $M > \sqrt{L_1 L_2}$

d) $M \leq \sqrt{L_1 L_2}$

$$V_1 = (L_1 - M)\frac{di_1}{dt} + M\frac{d}{dt}(i_1 - i_2)$$

$$V_2 = M\frac{d}{dt}(i_1 - i_2) + M\frac{di_2}{dt} - L_2\frac{di_2}{dt}$$

$$V_2 = M\frac{d}{dt}(i_1 - i_2) - (L_2 - M)\frac{di_2}{dt}$$



$$V_1 = (L_1 - M)\frac{di_1}{dt} + M\frac{d}{dt}(i_1 - i_2)$$

$$V_2 = M\frac{d}{dt}(i_1 - i_2) - (L_2 - M)\frac{di_2}{dt}$$

In the above equations for V_1 and V_2, the turns ratios are not considered.

The above model is applicable only for subtractive coupling.

REFLECTED IMPEDANCE

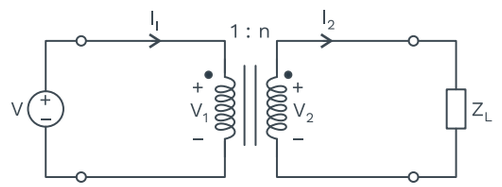


Fig. 9.12 Relating the Primary and the Secondary Quantities in an Ideal Transformer

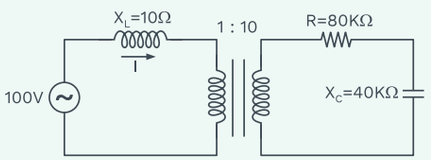
The input impedance, as seen from the source, is found from equations 4) and 6) as:

$$Z_{in} = \frac{V_1}{I_1} = \frac{V_2}{I_2}$$

From the figure 9.12, it is clear that $\frac{V_2}{I_2} = Z_L$ so the input impedance is related to the load impedance as:

Previous Years' Question

The following figure shows the connection of an ideal transformer with primary to the secondary turns ratio of 1 : 100. The applied primary voltage is 100(rms), 50 Hz, AC. The rms value of current I, in ampere is:



$$Z_{in} = \frac{Z_L}{n^2} \quad ...(7)$$

- "The above equation represents the input impedance (Z_{in}) which is referred as reflected impedance."
- Equation 7) represents that the load impedance is reflected to the primary side.

Elimination of the transformer by reflecting impedances and the sources from one side to the other:

There are two ways of analyzing the circuit shown in figure 9.13, one is by reflecting the secondary side impedance and the voltages to the primary side, and the other is by reflecting the primary side impedance to the secondary side.

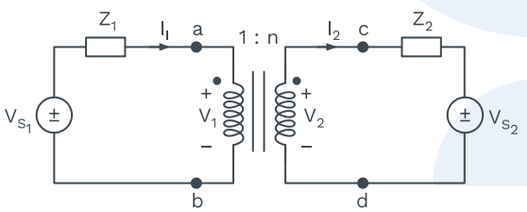


Fig. 9.13 Ideal Transformer Circuit to the Primary Side

Method-1:

The rule for the elimination of the transformer and reflecting the secondary circuit to the primary side is to divide the secondary impedance (Z_2) by n^2, divide the secondary voltage $\left(V_{s_2}\right)$ by n, and multiply the secondary current by n, as shown in figure 9.14.

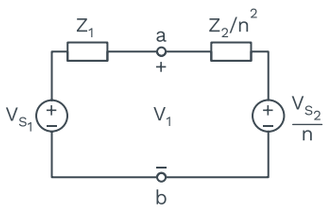
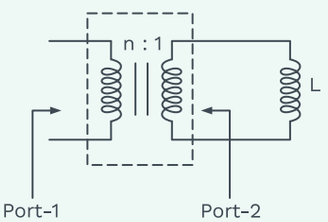


Fig. 9.14 Equivalent Circuit of Reflected Impedance by Method-1

Previous Years' Question

If an ideal transformer has an inductive load element at port-2 as shown in the figure below, the equivalent inductance at port-1 is_____.



a) nL **b)** n^2L

c) $\frac{n}{L}$ **d)** $\frac{n^2}{L}$

Method-2:

Reflecting the primary circuit to the secondary side:

The rule for the elimination of the transformer and reflecting the primary circuit to the secondary side by multiplying the primary impedance (Z_1) by n^2, multiply the primary voltage (V_{s_1}) by n, and divide the primary current by n, as shown in figure 9.15.

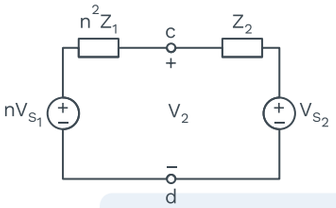
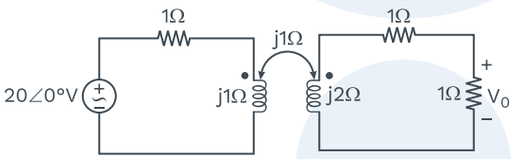


Fig. 9.15 Equivalent Circuit of Reflected Impedance by Method-2

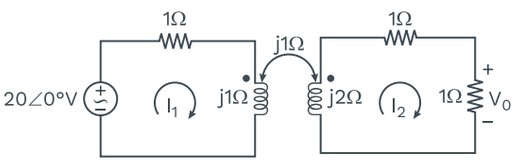
Example 8:

For the circuit shown in figure, find the output voltage (V_0).



Sol:

Writing mesh equation in primary and secondary circuits:



Mesh 1: $20\angle 0° - 1I_1 - j1I_1 + j1I_2 = 0$

$I_1(1 + j1) - j1I_2 = 20\angle 0°$ ...(i)

Mesh 2: $-j2I_2 - 1I_2 - 1I_2 + j1I_1 = 0$

$j1I_1 - I_2(2 + j2) = 0$ ...(ii)

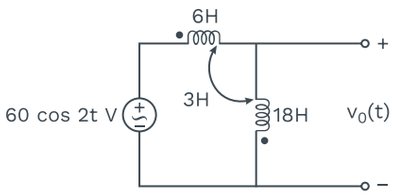
Solving equations i) and ii), we get:

$I_2 = 4.85\angle 14°A$

$V_0 = I_2(1) = 4.85\angle 14°$ V

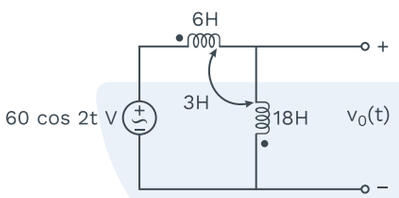
Example 9:

Obtain the output voltage $v_0(t)$ in the circuit.

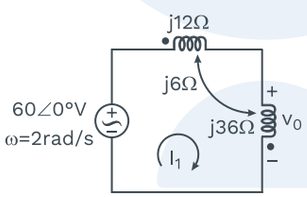


Sol:

Given,



Consider the circuit in the phasor domain,



$$60 - j12I_1 - j36I_1 + \underbrace{j6I_1 + j6I_1}_{\text{Due to mutual inductance}} = 0$$

$$j12I_1 + j36I_1 - j12I_1 = 60$$

$$j36I_1 = 60$$

$$I_1 = \frac{5}{3}\angle -90°\text{A}$$

$$V_0 = (j36 - j6)I_1$$

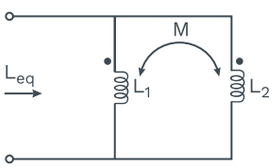
$$= j30 \times \frac{5}{3}\angle -90°$$

$$V_0 = 50\angle 0°\text{V}$$

$$v_0(t) = 50 \cos 2t \text{ V}$$

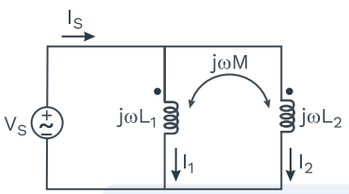
Example 10:

Two magnetically coupled coils are connected, as shown in the figure. The equivalent inductance L_{eq} is:



Sol:

Considering the circuit in the phasor domain



$I_s = I_1 + I_2$

$V_s = (j\omega L_1)\, I_1 + (j\omega M) I_2$ ...(i)

$V_s = (j\omega L_2)\, I_2 + (j\omega M) I_1$ ...(ii)

Substituting $I_1 = I_s - I_2$ into equation i)

$V_s = (j\omega L_1)(I_1 - I_2) + j\omega M I_2$

$V_s - j\omega L_1 I_s = I_2(M - L_1)j\omega$

$$I_2 = \frac{V_1 - j\omega L_1 I_s}{j\omega(M - L_1)} \quad \text{...(iii)}$$

Substituting $I_1 = I_s - I_2$ into equation ii)

$V_s = j\omega L_2 I_2 + (I_1 - I_2)j\omega M$

$V_s = j\omega(L_2 - M) I_2 + j\omega M I_1$

Substituting I_2 from equation iii)

$$V_s = j\omega(L_2 - M)\left[\frac{V_s - j\omega L_1 I_s}{j\omega(M - L_1)}\right] + j\omega M I_s$$

$$V_s\left[1 - \frac{j\omega(L_2 - M)}{j\omega(M - L_1)}\right] = I_s\left[\frac{\omega^2 L_1(L_2 - M)}{j\omega(M - L_1)} + j\omega M\right]$$

$V_s\,[j\omega M - j\omega L_1 - j\omega L_2 + j\omega M] = I_s(\omega^2 L_1 L_2 - \omega^2 L_1 M - \omega^2 M^2 + \omega^2 L_1 M)$

$$\frac{V_s}{I_s} = \frac{\omega^2\left(L_1 L_2 - M^2\right)}{j\omega(2M - L_1 - L_2)}$$

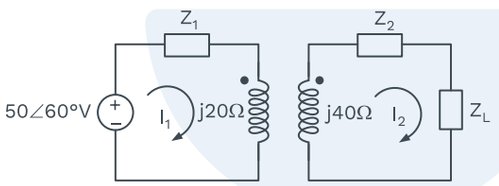
$$= \frac{j\omega\left(L_1L_2 - M^2\right)}{L_1 + L_2 - 2M}$$

$$\frac{V_s}{I_s} = j\omega L_{eq}$$

So, $L_{eq} = \dfrac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$

Example 11:

In the circuit of the figure, obtain the input impedance and the current (I_1) for the given network. If $Z_1 = 60 - j100\Omega$, $Z_2 = 30 + j40\Omega$ and $Z_L = 80 + j60\Omega$.



Sol:

Solving the circuit, we get,

$$Z_{in} = Z_1 + j20 + \frac{(5)^2}{j40 + Z_2 + Z_L}$$

$$Z_{in} = 60 - j100 + j20 + \frac{25}{j10 + j140}$$

$$Z_{in} = 60 - j80 + 0.14\angle -51.84°$$

$$Z_{in} = 60.09 - j80.11 = 100.14\angle -53.1° \text{ A}$$

$$I_1 = \frac{V}{Z_{in}} = \frac{50\angle 60°}{100.14\angle -53.1°}$$

$$I_1 = 0.5\angle 113.1°\text{A}$$

Example 12:

An ideal transformer has 50 turns on the secondary side, and it is rated at 2400/120V, 9.6 kVA.

Obtain:

a) the turn ratio

b) the number of turns on the primary side

c) the current ratings for the primary and secondary windings.

Sol:

a) According to the question, the given transformer is a step-down transformer.

$V_1 = 2400V > V_2 = 120V$

$$n = \frac{V_2}{V_1} = \frac{120}{2400} = 0.05$$

b) $$n = \frac{N_2}{N_1} \Rightarrow 0.05 = \frac{50}{N_1}$$

$$\text{or, } N_1 = \frac{50}{0.05} = 1000 \text{ turns}$$

c) $S = V_1 I_1 = V_2 I_2 = 9.6$ kVA

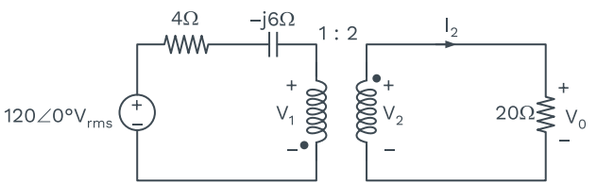
$$\text{Hence, } I_1 = \frac{9600}{V_1} = \frac{9600}{2400} = 4 \text{ A}$$

$$I_2 = \frac{9600}{V_2} = \frac{9600}{120} = 80 \text{ A}$$

$$\text{or, } I_2 = \frac{I_1}{n} = \frac{4}{0.05} = 80 \text{ A}$$

Example 13:

Determine a) the source current (I_1) b) the output voltage (V_0) and c) the complex power supplied by the source for an ideal transformer shown below.



Sol:

a) The 20Ω load impedance can be reflected to the primary side as:

$$Z_R = \frac{20}{n^2} = \frac{20}{4} = 5\Omega$$

Thus, $Z_{in} = 4 - j6 + Z_R = 9 - j6\Omega$

$Z_{in} = 10.82\angle -33.69°\Omega$

$$I_1 = \frac{120\angle 0°}{Z_{in}} = \frac{120\angle 0°}{10.82\angle -33.69°}$$

$I_1 = 11.09\angle 33.69°$ A

b) Since both the currents I_1 and I_2 leave the dotted terminals, so:

$$I_2 = -\frac{4}{n} I_1 = -5.545\angle 33.69°\text{A}$$

$$V_0 = 20\ I_2 = 110.9\angle 213.69°\text{V}$$

c) The complex power supplied is-

$$S = V_S I_1^* = (120\angle 0°)(11.09\angle -33.69°)$$

$$S = 1330.8\angle -33.69°\ \text{VA}$$

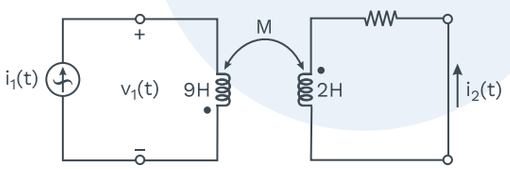
Example 14:

In the circuit given below, the 2H inductor is short-circuited. The currents $i_1(t)$ and $i_2(t)$ are:

$i_1(t) = 2e^{-t}$ A, $\quad t \geq 0$

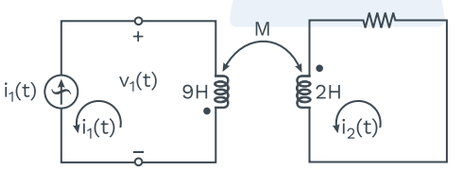
and $i_2(t) = 4e^{-t}$ A, $t \geq 0$

The voltage $v_1(t)$ for $t > 0$ is expressed as:



Sol:

First, we obtain the value of mutual inductance M.



KVL in the secondary loop

$$0 = 2\frac{di_2(t)}{dt} - M\frac{di_1(t)}{dt}$$

$$0 = 2\frac{d}{dt}\left(4e^{-t}\right) - M\frac{d}{dt}\left(2e^{-t}\right)$$

$$0 = 8\left(-e^{-t}\right) - M\left(-2e^{-t}\right)$$

put t = 0, we get:

$0 = -8 + 2M$

or $M = 4$

Applying KVL in primary loop

$$v_1(t) = 9\frac{di_1(t)}{dt} - M\frac{di_2(t)}{dt}$$

$$= 9\frac{d}{dt}\left(2e^{-t}\right) - 4\frac{d}{dt}\left(4e^{-t}\right)$$

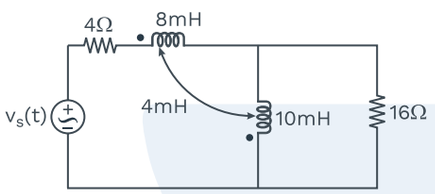
$= 9(-2e^{-t}) - 4(-4e^{-t})$

$= -18e^{-t} + 16e^{-t}$

$v_1(t) = -2e^{-t}$ V

Example 15:

For the given circuit, $v_s(t) = 50\cos(1000t)$V. The average power delivered to the 16Ω resistor will be _____ watt.



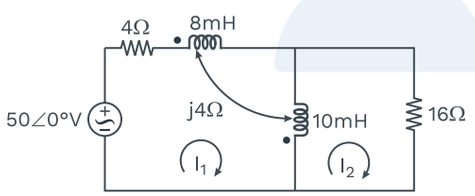
Sol:

Considering the circuit into phasor domain

$8mH \Rightarrow j\omega L_1 = j(10^3)(8\times10^{-3}) = j8\Omega$

$10mH \Rightarrow j\omega L_2 = j(10^3)(10\times10^{-3}) = j10\Omega$

$4mH \Rightarrow j\omega M = j(10^3)(4\times10^{-3}) = j4\Omega$



Mesh I:

$50 = 4I_1 + j8I_1 + j10(I_1 - I_2) - j4(I_1 - I_2) - j4I_1$

$50 = I_1(4 + j8 + j10 - j4 - j4) + I_2(-j10 + j4)$

$50 = I_1(4 + j10) - j6I_2$ .....i)

Mesh II:

$0 = j10(i_2 - I_1) + 16I_2 + j4I_2$

$0 = I_1(-j10 + j4) + I_2(j10 + 16)$

$j6I_1 = (16 + j10)I_2$

$$I_1 = \left(\frac{16 + j10}{j6}\right)I_2$$

Substituting I_1 into equation i)

$$50 = \left(\frac{16 + j10}{j6}\right)(4 + j10)I_2 - j6I_2$$

$(j6)\ (50) = (-36 + j200 + 36)I_2$

$I_2 = 1.5\angle 0°A$

Average power absorbed by 16Ω resistor

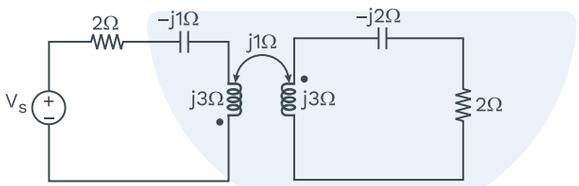
$$P_{16\Omega} = \frac{1}{2}|I_2|^2 R$$

$$= \frac{1}{2}(1.5)^2\,(16)$$

$P_{16\Omega} = 18\ W$

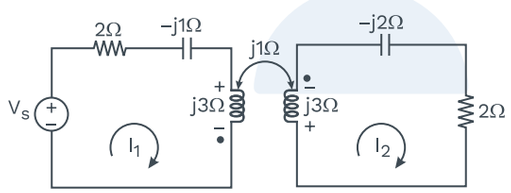
Example 16:

Impedance seen by the source $V_s = 24\angle 0°$ in the network shown in figure is:



Sol:

Applying KVL in both loops (considering mutual inductance)



In secondary loop

$$-(-j2I_2) - 2I_2 - j3I_2 - \underbrace{j1I_1}_{\text{mutual voltage}} = 0$$

$(j3 - j2 + 2)I_2 + j1I_1 = 0$

$I_2(j1 + 2) + jI_1 = 0$

$$I_2 = -\frac{j1}{2 + j1}I_1$$

$$I_2 = \frac{-j(2 - j)}{5}I_1$$

In primary loop

$$V_s - 2I_1 - (-jI_1) - j3I_1 - \underbrace{j1I_2}_{\text{mutual voltage}} = 0$$

$V_s - 2I_1 + j1I_1 - j3I_1 - j1I_2 = 0$

$V_s = (2 + j2)I_1 + j1I_2$

Substituting I_2 from equation i), we have

$$V_s = (2 + j2)I_1 + j\left[\frac{-j(2 - j1)}{5}\right]I_1$$

$$= \left(\frac{10 + j10 + 2 - j}{5}\right)I_1$$

$$= \left(\frac{12}{5} + \frac{9}{5}j\right)I_1$$

$V_s = (2.4 + 1.8j)I_1$

The impedance seen by the source is:

$$Z_s = \frac{V_s}{I_1}$$

$Z_s = (2.4 + 1.8j)\Omega$

Chapter Summary



1) The two coils are considered to be as mutually coupled, if the magnetic flux (ϕ) derives from coil one passes through the other. The mutual inductance between the two coils are given as: $M = K\sqrt{L_1\ L_2}$.

Where K is the coupling coefficient, $0 < K \leq 1$.

a) If $K < 1$; loose coupling (Under coupling circuit).

b) If $K = 1$ i.e. $M = K\sqrt{L_1\ L_2}$; (Critical coupling)

c) $K > 1$; tight coupling (Over coupled circuit)

2) If the voltage and current of coil (1) are v_1 and i_1 respectively and the voltage and current in coil (2) are v_2 and i_2 respectively, so the equation of voltages are:

$$v_1 = L_1\frac{di_1}{dt} + M\frac{di_2}{dt} \text{ and } v_2 = L_2\frac{di_2}{dt} + M\frac{di_1}{dt}$$

Thus, the above equation represents the voltage induced in a coupled coil which consists of a self-induced voltage and a mutual voltage.

3) To determine the polarity of an induced emf, dot convention is used at the terminals of a coil.

4) The equation of energy stored in a two coupled coils is given as:

$$E_{stored} = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$$

5) A four-terminal device which contains two or more magnetically coupled coils is called as the transformer. The transformer is used to change the current, voltage and the impedance level in a network.

6) The coils of a linear transformer are wound on a magnetically linear network. The network is replaced by an equivalent (T) or (Π) network.

7) A transformer with unity coupling coefficient ($K = 1$) having no leakage flux and has infinite inductance ($L_1, L_2\ M \to \infty$) is termed as an ideal transformer.

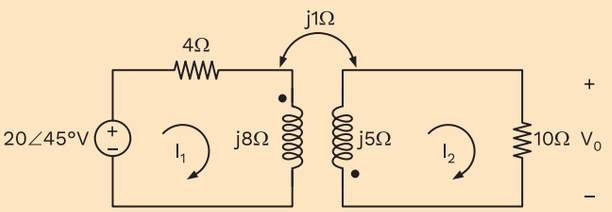
8) For an ideal transformer $V_2 = nV_1$, $I_2 = \frac{I_1}{n}$, $S_1 = S_2$, $Z_R = \frac{Z_L}{n^2}$ where $n = \frac{N_2}{N_1}$ is the turns ratio, N_1 = Number of primary winding turns and N_2 = Number of secondary winding turns.

9) A transformer with a single winding which is both common to the primary section and the secondary section is known as an autotransformer.

10) The transformers are used as an electrical isolating device, and impedance matching device in electronics.

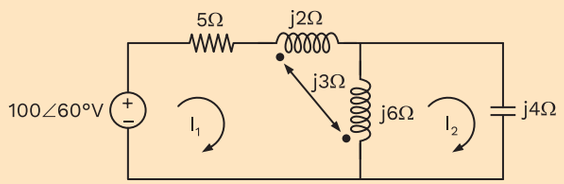
PRACTICE QUESTIONS

Q1 **Determine the voltage V_0 in the circuit in the figure.**



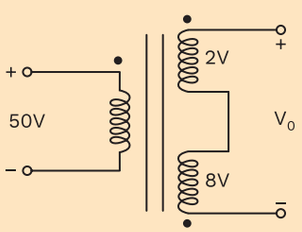
a) $20\angle -135°$V **b) $20\angle 135°$V** **c) $20\angle 0°$V** **d) $25\angle 0°$V**

Q2 **For the circuit shown in figure, the value of phasor currents I_1 and I_2 are:**



a) $I_1 = 17.889\angle 86.57°$ A, $I_2 = 26.83\ \angle 86.57°$ A
b) $I_1 = 17.889\ \angle -86.57°$ A, $I_2 = 26.83\ \angle -86.57°$ A
c) $I_1 = 17\angle -80°$ A, $I_2 = 17\ \angle -80°$ A
d) $I_1 = 17\ \angle 80°$ A, $I_2 = 17\ \angle 80°$ A

Q3 **For a three winding transformer, as shown in the figure, the value of the output voltage is:**



a) – 10 V **b) + 10 V** **c) + 8 V** **d) – 6 V**

Q4 **The transformer which is used as an isolation device is:**

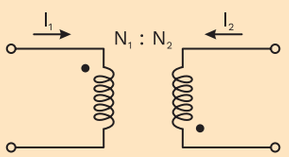
a) Linear transformer **b) Ideal transformer**

c) Auto transformer **d) All of the above**

Q5 **The two coils having inductance L_1 = 4H, L_2 = 4H, and M = 2H, then the coefficient of coupling is:**

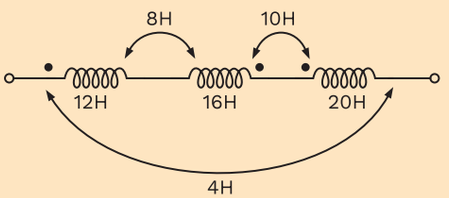
a) 4 **b) 8** **c) 2** **d) 0.5**

Q6 **The ideal transformer has $\frac{N_2}{N_1} = 10$, the ratio $\frac{I_2}{I_1}$ is:**



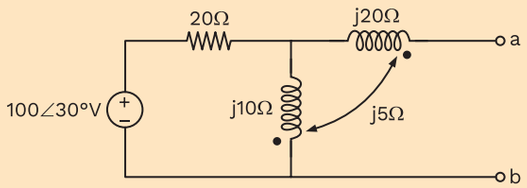
a) 0.1 **b) 10** **c) – 0.1** **d) – 10**

Q7 **For the coupled coils in the figure, calculate the total inductance.**



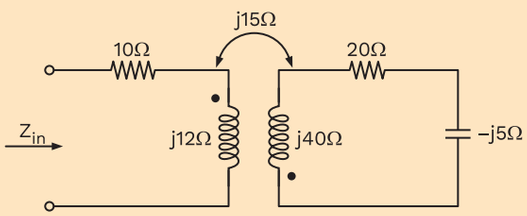
a) 12 H **b) 5 H** **c) 9 H** **d) 10 H**

Q8 **Find the Norton equivalent for the circuit in the figure at terminals a – b.**



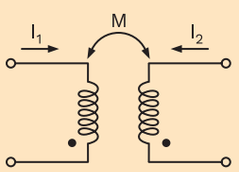
a) $Z_N = (1 + j19.5)\Omega$, $I_N = 1.404\ \angle 9.44°A$
b) $Z_N = (1 - j19.5)\Omega$, $I_N = 1.404\ \angle 9.44°A$
c) $Z_N = (3 + j50)\Omega$, $I_N = 1.5\ \angle 10°A$
d) $Z_N = (1 - 2j)\Omega$, $I_N = 2\ \angle 3.5°A$

Q9 **For the network shown below, obtain the input impedance of an air-core transformer.**



a) $(3 + j40)\Omega$
b) $(12.769 + j7.154)\Omega$
c) $(13 + j5)\Omega$
d) $(30 + j52)\Omega$

Q10 **The polarity of the mutual voltage for the two magnetic coupled coils as shown below is:**



a) Positive
b) Negative
c) Both positive and negative
d) None of these

Answer Key:

1)	**a)**	**2)**	**a)**	**3)**	**d)**	**4)**	**b)**	**5)**	**d)**
6)	**a)**	**7)**	**d)**	**8)**	**a)**	**9)**	**b)**	**10)**	**a)**